

10/6/26

Python Programming Assignment-2

1) Write a program to display:

- value -

```
print("Jay Ganesh...")
```

- Type -

```
x = 11
```

```
print(type(x))
```

```
# <class 'int'>
```

- Memory address -

```
x = 11
```

```
print(id(x))
```

```
# 14071911080024
```

2) What is difference between:

```
a = 10
```

```
b = 10
```

and

```
a = [10]
```

```
b = [10]
```

Explain using id().

Ans :

The difference between the above two cases depends on how Python manages memory for immutable versus mutable objects.

Case 1: $a = 10$ & $b = 10$

In this case $id(a)$ and $id(b)$ will be returned the same memory address and data type is integer. So integers are immutable since value of both variables are constant ~~or same~~ same constant it points them both to the same object in the memory therefore the will be same.

Case 2: $a = [10]$ & $b = [10]$

In this case $id(a)$ and $id(b)$ will return different memory address because given values ~~are~~ is list. Lists are mutable. Everytime we declare a new list using square brackets $[]$ this occurs a new object in memory even though the contents are same, it identifies as two different containers so they have different memory address / ids.

3) What does the $id()$ function return? Is the value returned by $id()$ same for two variables holding the same value?

Ans :

The $id()$ function returns the memory address of the value. And yes the value returned by $id()$ of two variables holding the same variable is same.

4) What is the purpose of `getsizeof()`? Why is memory size different for different datatypes?

Ans: The purpose of `getsizeof()` function is it returns the size of memory of an object in bytes.

Memory size different for different datatype because every data type stores different amounts of information, so it needs different memory.

For example -

<code>x = 10</code>	<code># int</code>
<code>y = 10.5</code>	<code># float</code>
<code>z = "Hello"</code>	<code># string</code>

- int stores a whole number
- float stores a decimal number which require a more complex format
- string stores multiple characters, so it needs space for each character.

5) Predict the output:

`a = 10`

`b = 10`

`print(id(a) == id(b))`

Explain why this happens.

Ans: The output of the above program is 'True'. This is because `id(a)` & `id(b)` will return same memory address because the both values are same and also the datatype is same i.e., integer. That's why the output is 'True'.

Q) Write a program that accepts two numbers from user and prints their:

- Addition
- Subtraction
- Multiplication
- Division

("05" + "01") trying
(05 + 01) trying

two digit

0501

08

Ans :- Program :-

```
x = int(input("Enter first number: "))
y = int(input("Enter second number: "))
```

```
print("\n--- Displaying the results ---")
```

Addition

```
Addition = x + y
```

```
print(f"Addition is : {x} + {y} = {Addition}")
```

Subtraction

```
Subtraction = x - y
```

```
print(f"Subtraction is : {x} - {y} = {Subtraction}")
```

Multiplication

```
Multiplication = x * y
```

```
print(f"Multiplication is : {x} * {y} = {Multiplication}")
```

~~# Division~~

Division :

Division

if $y \neq 0$:

Division = x / y

print(f"Division is: {x} / {y} = {Division}")

else:

print("Error: Division by zero is not allowed")

7) Why does the input() function always return a string?
How can you convert it to another data type?

Ans: The name says it all. The input() function is used so that the user can type any input so it can be any alphanumeric characters that's why it is considered as a string.

We can convert it to another datatype by just declaring a datatype before the input() function.
for ex:-

```
x = int(input("Enter a number: "))
```

8) Predict the output:

```
x = input("Enter number: ")
print (type(x))
```

<class 'str'>

This is because input() function by default considers its datatype as string until any sep developer assign any datatype before input function.

9) Write a program to take user's name and age and display:

Hello <name>, you will turn <age+1> next year.

Ans: Program:

```
name = str(input("Enter your name: "))
age = int(input("enter your age: "))
```

```
print(f"Hello {name}, you will turn {age+1} next year.")
```

output:

Enter your name: Sahil Patil

Enter your age: 24

Hello Sahil Patil, you will turn 25 next year.

10) What will be the output:

```
print("10" + "20")  
print(10 + 20)
```

output:

1020

30

The difference is the quotes (" "). Anything written in quotes Python ~~considered~~ considers it as a String. That's why output is 1020. at

And in next case it ~~con~~ its datatype is integers so it ~~be~~ performs mathematical operation i.e., Addition and the output is 30.